

AVULA PUNITH KUMAR REDDY

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Professional Summary

Dynamic Mechanical Design Engineer with 1+ years of hands-on experience in CAD modeling, product design, and engineering analysis at Bharat Electronics Limited. Expert in SolidWorks, AutoCAD, and GD&T for developing precision components and assemblies that meet stringent manufacturing standards. Demonstrated success in applying DFM principles to optimize designs, resolve production challenges, and contribute to cost-effective solutions while ensuring compliance and quality excellence.

Technical Skills

- **CAD/Modeling:** SolidWorks, AutoCAD, CATIA V5, Siemens NX
- **Design Skills:** 3D Modeling, Assembly Design, GD&T, Sheet Metal, Welding
- **Analysis & Simulation:** SolidWorks Simulation, ANSYS (Basic)
- **Tools:** MS Office Suite, ERP Systems (SAP)

Experience

Bharat Electronics Limited

Mechanical Design Engineer

Hyderabad, Telangana

Feb 2025 – Present

- Developed detailed 3D models and 2D manufacturing drawings for mechanical components using SolidWorks and AutoCAD, ensuring 100% compliance with industry standards.
- Collaborated with the manufacturing team to resolve production bottlenecks.
- Applied Geometric Dimensioning and Tolerancing (GD&T) to ensure precision and interchangeability of complex parts in multi-component assemblies.
- Maintained and updated Bill of Materials (BOM) for multiple product lines within the ERP system, ensuring procurement accuracy.
- Conducted static load analysis using SolidWorks Simulation to validate design integrity under operating conditions.

Education

JNTUH-UCES

Bachelor of Technology in Mechanical Engineering; CGPA: 6.9/10.0

Sangareddy, Telangana

2024

Sri Chaitanya Jr kalashala

Intermediate; Percentage: 91.4%

Hyderabad, Telangana

2018 - 2020

Unique High School

10th Class; CGPA: 9.7/10.0

Hyderabad, Telangana

2018

Projects

Bladeless Fan using Ion Wind | *SolidWorks, Electrohydrodynamics*

- Designed and prototyped a bladeless fan utilizing Ion Wind (Ion thrust) to generate airflow through high-voltage corona discharge, eliminating the need for moving parts.
- Engineered and optimized emitter and collector electrode geometries in SolidWorks to maximize airflow velocity and structural stability.
- Conducted experimental analysis to quantify the relationship between voltage input and airflow output.
- Integrated high-voltage power supplies and fabricated a specialized housing using FDM 3D printing (PLA) to ensure robust electrical insulation, structural integrity, and user safety.